Week 11 – Lab Assignment

CSC 3320 – System-level Programming

Thursday April 2nd, 2026

Introduction

Welcome to the 11th programming lab of CSC 3320! Today, we will be covering the following topics:

1. Pointers
2. Dynamic Memory Allocation

Lab Policies

- Attendance is mandatory.
- Labs must be completed **individually.**
- TAs are here to help you. Ask them for help!
- Lab assignments are due at midnight on the day of your lab.

Comments

The lab assignment requires the inclusion of comments to enhance code readability and understanding. Specifically, a block comment at the beginning of the C file is required. Your block comment should include the following:

- The program name
- The author's name (your name)
- A description of the program's overall purpose

Deliverables:

1. The C Code for your program. (.c file).
2. A screenshot of the output in the Terminal.

If you have any questions, please do not hesitate to ask your TA.

Program: Number sequences – memory allocation

Complete the following functions to generate different sequences of numbers:

- `int* Fibonacci(int n)`
  - Create an array of integers and size n
  - Fill in the array with the first n Fibonacci numbers. The Fibonacci sequence begins with 0 and then 1 follows. All subsequent values are the sum of the previous two, ex: 0, 1, 1, 2, 3, 5, 8, 13.
  - Return the array
- `int* Squares(int n)`
  - Create an array of integers and size n
  - Fill in the array with the squares of 1 to n (inclusive). Ex: 1, 4, 9, …, n^2
  - Return the array
- `int* Concatenate(int* array1, int size1, int* array2, int size2)`
  - Create an array of integers and size = size1 + size2
  - Fill in the array with the elements of array1 followed by the elements of array2
  - Return the array

main() reads the size of the Fibonacci and the squares sequences and calls the three functions above to create the arrays. Then main() calls PrintArray() provided in the template to print the arrays.

Ex: If the input is:

```
6 4
```

the output is:

```
0 1 1 2 3 5
1 4 9 16
0 1 1 2 3 5 1 4 9 16
```

Deliverables

For today's lab, you must upload the C program code for your grade management program and its output in the terminal on iCollege. Please name your C code and screenshot as follows:

- C Files
  - lastname_firstname_filename.c
  - For example: **turing_alan_number_sequences.c**
- Screenshots
  - lastname_firstname_filename.png
  - For example: **turing_alan_ number_sequences.png**